

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination October/November 2016

PH222 PHYTOCHEMISTRY-I

Date: 10-11-2016 , Thursday Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

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| Q 1 | Attempt all questions. Each question is of one mark. | 20 |
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1. Which of the following is obtained as racemic compound?
A. Morphine
B. Atropine
C. Reserpine
D. Quinine
 2. Which of the following is official source for isolation of diosgenin?
A. *Dioscorea flouribanda*
B. *Dioscorea deltoidea*
C. *Dioscorea composite*
D. All of the above
 3. Opium alkaloids occur in combination with
A. Lysergic acid
B. Meconic acid
C. Quinic acid
D. Chlorogenic acid
 4. _____ is complex formed on treating anthraquinones with alkali.
A. Phenolate ion
B. Quinones
C. naphthoquinones
D. kinetin
 5. Heroin can be differentiated from morphine by presence of _____ group.
A. Ethyl
B. Methyl
C. Acetyl
D. Dihydro
 6. Steroidal saponins are used as starting material for the synthesis of _____.
A. Vitamin A
B. Vitamin C
C. Vitamin D

7. Find the phytoconstituent with analgesic property
 - D. Vitamin B6
 - A. Papaverine
 - B. Morphine
 - C. Atropine
 - D. Reserpine
8. Kedde's reaction gives red violet color with _____
 - A. Quinoline Alkaloid
 - B. Phenanthrene Alkaloid
 - C. Cardiac Glycoside
 - D. Anthraquinone Glycoside
9. Quercetin is the aglycone obtained from _____
 - A. Sennoside
 - B. Rutin
 - C. Digitoxin
 - D. Aloin
10. Rubane ring is present in _____
 - A. Reserpine
 - B. Rutin
 - C. Papaverine
 - D. Quinine
11. _____ is synthesized in plant through shikkinic acid pathway
 - A. Dioscine
 - B. Digitoxigenin
 - C. Gitoxigenin
 - D. Ephedrine
12. Sennosides are used as _____
 - A. Anticancer
 - B. Laxative
 - C. Cardiotonic
 - D. Antihypertensive
13. _____ reagent is used for derivatization of ephedrine on TLC.
 - A. Dragendorff's
 - B. Modified dragon droff's
 - C. Ferric chloride
 - D. Ninhydrine
14. Shinoda tests is for _____
 - A. cardiac steroid
 - B. triterpenoidal saponin
 - C. Flavonoid
 - D. glycosylated lignin
15. _____ is used in treatment of cancer
 - A. Vincristine
 - B. Rutin
 - C. Atropine
 - D. Quinine
16. Which of the following constituents is not used for direct medication.
 - A. Diosgenin
 - B. Quinine
 - C. Rutin

D. Morphine

17. Which of the following pseudo alkaloid is identified by potassium chlorate and ammonia.
A. connessine
B. caffeine
C. aconitine
D. protoveratrine A
18. Rutin is known as vitamin _____ and is used to reduce _____.
A. A, macular degeneration
B. P, capillary fragility
C. E, oxidation
D. R, lipid peroxidation
19. Which one of following does possess deoxy sugar?
A. Digitoxin
B. Aloin
C. Sennoside
D. Rutin
20. _____ is lysergic acid derivative
A. Quinine
B. Morphine
C. Ergotamine
D. Ephedrine

SECTION – II

- Q 2 Attempt any **TWO** of the following
- A Define secondary metabolites and draw general biogenetic pathway for Alkaloids. 05
- B Enumerate methods for basic biogenetic pathway elucidation. Write in detail about radio isotop tracer technique. 05
- C Describe objectives of performing biogenetic pathways and chemistry of secondary metabolites by giving suitable examples. 05
- Q 3 Attempt any **TWO** of the following
- A Describe General Procedure for Isolation of Alkaloids from plant and write important steps involved in isolation of morphine from opium with justification. 05
- B State Identification tests and describe isolation procedures of Quinine 05
- C Prepare a Note on : Application of Hofmann Exhaustive Methylation in Structure Elucidation of Natural Products 05

SECTION – III

- Q 4 Attempt any **FOUR** of the following
- A Justify why Indian podophyllum is medicinally important source than American. Prepare a note on Lignan obtained from podophyllus with special reference to isolation and application 05
- B Give details on stereoisomerism on Cinchona and Ephedra alkaloids. 05
- C Write a note on General Chemistry of Lignans 05
- D Write a note on application of knowledge of stereochemistry in natural products 05
- E Describe stereochemistry of Atropine 05
- F Write a Note on : Emde Degradation 05

- Q 5 Attempt any FOUR of the following
- A Differentiate : Cardenolide and Bufadienolide (Write 5 major Differences) 05
- B Prepare a note on isolation and application Glycerrhizin 05
- C Write outline of isolation procedures for diosgenin from its official sources. 05
- D What is 16-DPA, write how it is synthesized from diosgenin. Write further conversion of 16-DPA in to commercially utilized products. 05
- E Write in Detail: Isolation of Digoxin and medicinal applications of cardiac glycosides 05
- F Describe major steps involved in isolation of Sennoside and mechanism, through which it acts as medicinal agent 05